Psychology of Learning: Revision Notes

NPTEL · IIT Kharagpur · Prof. Atasi Mohanty. All 12 weeks in lecture order, written from the 60 lecture transcripts and slides.

How to use these. Bold words are the keywords answer options tend to use. Orange boxes are exam tips: traps, “which is NOT” answers, and places where the lecture differs from textbooks. NPTEL marks by the lecture, so these notes follow it. Pair them with the “Who Gave What” sheet for names.

Week 1 · Nature and Principles of Learning

Defining learning

- Schools in order: **structuralism** → **functionalism** → **behaviorism** → **cognitivism** → **constructivism**
- Learning is central to psychology and "can be defined in many ways"
- Learning = **"relatively permanent change in behavior that results from experience"**; the **"contiguous causal effect of experience on behavior"**
- Learning needs an **observable change in behavior** within the organism's lifetime, caused by **regularities in the environment**. It is an effect attributed to an element in the **environment**
- **Cognitive definitions**: a change in **mental constructs** (knowledge, representations, associations)
- Learning changes the **physical structure of the brain**

Exam tip: "a hidden change in behavior" = False. A skeleton is NOT a mental construct.

Principles and history

- Learning depends on the environment, reinforcement/feedback, **association** with past experience, and **cultural and social context**
- Learning styles: **visual, auditory, kinesthetic**. **Both thoughts and emotions** shape learning
- **Watson**: "**Psychology as the Behaviorist Views It**"; **Little Albert** conditioned to **fear** a white rat. Behaviorism dominated the early **20th century**
- The field covers motivation, memory, retention and forgetting, NOT genetic disorders
- **Conscious incompetence** is valuable **at the start** of learning
- Effective learning gives **more connected knowledge**, wider strategies, a reflective approach, **positive emotions**, and a **vision of the future self as a learner**

Exam tip: "sad/anxious feelings" and "disconnected knowledge" are NOT outcomes of effective learning.

Principles of how students learn

- **Conceptual growth** = adding to existing knowledge; **conceptual change** = revising it
- Transfer to new contexts is "not spontaneous"; it must be facilitated
- **Deliberate practice** = **attention, rehearsal, repetition** (NOT forgetting)
- Feedback must be **clear, explanatory, timely**
- **Intrinsic** motivation beats extrinsic, but the two complement each other
- **Mastery goals** make students persist and process deeply; **performance goals** do not
- Best goals: **proximal**, specific, **moderately challenging**. They make students **intermediate risk takers**

Types of learning

- **Non-associative** learning = a **decrease** in response to a repeated stimulus
- **Habituation** = decrease (getting used to a ticking clock); **sensitization** = increase
- **Associative** learning links two events. **Instrumental conditioning** is associative
- **Meaningful learning** (**Ausubel**): new information is linked to prior concepts
- **Cooperative learning** (**Dewey**): "exchanges of information" and a "confluence of ideas"
- Its three elements: **heterogeneous groups, positive interdependence, individual responsibility**
- **Emotional learning** is used in therapy (**systematic desensitization**)
- **Observational learning** (**Bandura**); **discovery learning** (**Bruner**); **rote learning** (repetition)

Exam tip: the lecture describes sensitization as drug- or stimulus-induced; MCQs key habituation = decrease, sensitization = increase.

Week 2 · Major Theories of Learning

Behaviorism

- Pioneers: **Watson, Skinner, Pavlov** (NOT Freud). S–R approach based on animal experiments
- Behaviorists hold that all behaviors **can be unlearned**. Watson saw psychology as **experimental**
- Three types of learning in **behavioral psychology**: **classical conditioning, operant conditioning, observational learning**
- Classical conditioning (**Pavlov**) timing types:
- **Delayed**: the US starts while the CS is still present
- **Trace**: the CS ends before the US begins
- **Simultaneous**: CS and US together (the strongest)
- **Backward**: the **US comes before the CS**
- **Stimulus discrimination** = responding to certain stimuli but not others
- **Higher-order conditioning** is weaker than first-order (True)
- Operant conditioning (**Skinner**): the four consequences
- **Positive reinforcement**: praise, compliments
- **Negative reinforcement**: escaping or avoiding something (a shock)
- **Positive punishment**: adding a penalty
- **Negative punishment**: removing something (timeout)
- **Shaping** = gradually changing the quality of a response
- Observational learning (Bandura) needs no reinforcement. Steps: **attention, retention, production, motivation**

Cognitivism

- Psychology = **psyche + logos**. **Wundt** = father of modern psychology; first lab at **Leipzig, 1879**
- Cognitivists said behaviorists ignored **thinking**. The mind is an **information processor**, like a computer
- Cognitivism is the mainstream for research on learning designs
- The shift to cognitivism came from **Tolman, Bruner and German Gestalt** (all of them)
- **Gestalt**: the whole is more than the sum of its parts; insight
- **IQ** = mental age ÷ chronological age × 100

Intelligence theories

- **Gardner** listed 7 intelligences: linguistic, logical-mathematical, **spatial**, bodily-kinesthetic, musical, interpersonal, **intrapersonal** (self-knowledge)
- He later added **naturalistic** and **existential**
- **Sternberg** triarchic theory (**1985**): analytical, creative, practical. An alternative to **g**, which covers only **academic** intelligence
- **PASS** (J.P. Das et al., 1994) = **Planning, Attention-Arousal, Simultaneous, Successive**. Its test is the **CAS = Cognitive Assessment System**
- **BBL = Brain-Based Learning**: learning in a **neuro-physiological** context
- **Guilford**: 180 abilities. Products: **units, classes, relations, systems, transformations, implications**
- **Gagne**: Conditions of Learning = **nine** Events of Instruction
- **Mastery learning** steps: pre-assessment, group initial instruction, formative assessments, corrective instruction, parallel formative assessments

Exam tip: "symbolic" is a Guilford content, NOT a product.

Constructivism

- Reality is determined by **the learner's experiences**; students bring **unique experiences**
- Types: **cognitive** (Piaget), **social** (**Vygotsky**; adds peer influence), **radical**. There is no "biological" type
- **Piaget** stages: **sensorimotor**, **preoperational**, **concrete operational**, **formal operational**

- **Vygotsky**: the **zone of proximal development** is the gap between what a learner does alone and what they can do with help
- **Reciprocal teaching** = dialogue between a teacher and a small group

Exam tip: "post-operational" is NOT a Piaget stage.

Week 3 · Memory and Cognition

Memory basics

- Memory can **both preserve and recover** information. Memories form through changes in **neurons**
- Connections between nerve cells = **synapses**
- Three processes: **encoding**, **storage**, **retrieval** (imagination is NOT one)
- Encoding changes information into **a usable form** to create a **new memory**
- Retrieval brings stored memories into conscious awareness
- Retrieval = recall = recollection: calling back information in response to a cue
- **Cognition**: "mental processes relating to the acquisition, storage, manipulation and retrieval of information"
- Domains of cognition include attention, memory, intelligence and **judgment** (NOT aggression)
- People with **social cognition** deficits struggle to **read facial expressions**
- Information-processing view: information is processed in **stages**; **there are limits** at each stage; the system is **interactive**; it is analogous to a computer

Exam tip: "no limit on processing" is the false assumption.

Atkinson and Shiffrin model

- **Sensory register**: very large capacity, **1–3 seconds**
- **STM**: **7±2** chunks, **20–30 seconds**. **Rehearsal** prevents quick loss from STM
- **Maintenance rehearsal** = rote repetition; **elaborative rehearsal** links to LTM
- Children rehearse on their own from about age **7**
- **LTM**: unlimited capacity, permanent, stored as schemata
- **Dual coding** = verbal + visual. Forgetting: encoding failure, retrieval failure, interference, decay, brain damage

Working memory and LTM types

- **Baddeley**'s multicomponent model:
- **Central executive**: controls the system
- **Visuo-spatial sketchpad**: visual and spatial information
- **Phonological loop**: verbal information
- **Episodic buffer**: the interface with episodic LTM
- **Multimodal** presentations enhance working-memory capacity
- **Declarative** memory = **episodic** + **semantic**
- **Episodic** = events reported from a person's life; **autobiographical episodic** = specific episodes of one's life
- **Procedural** = implicit "know how", not consciously recalled
- **Levels of processing** (**Craik and Lockhart**): **structural** (physical qualities) → **phonemic** → **semantic** (deep). Deeper means better recall
- **Neural network model**: capacity depends on synapse complexity, sparseness of representations, spatial/temporal correlations and retrieval (all)

Exam tip: "language repository" is NOT a working-memory component.

Cognitive load

- Types: **intrinsic** (task), **extraneous** (presentation), **germane** (schema-building)
- Three measures: **mental load**, **mental effort** (the "cognitive capacity actually allocated"), **performance**. Physical effort is NOT one
- Causes of load: subject, task, environment and how they interact. Schemas reduce working-memory load
- Improve memory: write it down, attach meaning, repeat, chunk, self-test, take a mental picture

Exam tip: "over-exerting yourself" does NOT help memory.

Week 4 · Thinking and Cognition

Thinking

- Thinking = "consciously generating and manipulating thoughts". It is a **higher mental process** and **sub-vocal talking**
- We think in verbal symbols, mathematical symbols, gestures and movements (all)
- Three main types:
- **Creative**: innovative, unconventional, useful ideas
- **Divergent**: many answers to open-ended problems
- **Symbolic**: mental representations, as in children's **imaginative play**
- **Convergent** thinking is NOT one of the three
- **Concepts** organize complex phenomena into simpler categories
- **Prototypes** are representative items of a concept
- Mental images use all senses. A mental representation can be a word, image or sound
- **Cognitive architecture** = a theory about the **structures** that create a mind

Exam tip: "thinking is only done in verbal symbols" is false.

Bloom's taxonomy

- A hierarchy for categorizing **learning objectives**. Domains: **cognitive, affective, psychomotor**
- **Cognitive** (1956, revised 2001): **remember** (lowest) → understand → apply → analyze → evaluate → **create**
- **Affective** (1964): receiving → responding → valuing → organizing → characterizing
- **Psychomotor** (**skilled behavior**): imitation → manipulation → precision → articulation → naturalization
- **Factual** knowledge = terminology and specific details; **conceptual** = categories and principles; **procedural** = how-to
- **Metacognitive** knowledge = knowledge about one's own knowledge

Metacognition

- "**Thinking about thinking**"; crucial for independent learning (it does NOT make you dependent)
- Two parts: **knowledge** of cognition and **regulation** of cognition (planning, monitoring, evaluating)
- Skills: knowing your limits, **self-monitoring**, **skimming** past subheadings of unimportant information

Critical and reflective thinking

- **Critical thinking** = "the ability to collect and analyze information to come to a conclusion"
- **Open-mindedness** gives an **unbiased conclusion**
- **Self-regulation** sets aside personal biases
- **Reflective thinking** grows through self-questioning, concept maps, reflective discussions and **negotiated learning**

Exam tip: "avoiding reflective discussions" is NOT a way to build reflective thinking.

Creativity and problem solving

- Creative thinking = "**think out of the box**"; any skill **anybody can develop**
- Stages: **preparation** (the inspiration phase) → **incubation** → **insight** ("eureka"/"aha") → **evaluation** → **elaboration**
- **Trial and error** = try until one solution works
- **Algorithm** = a rule that **guarantees** a solution
- **Analogy** = reuse a past solution
- In the traditional approach, the student is **passive**
- **Concept maps** (**Ausubel**): two nodes joined by a labeled arrow = a **proposition**

Exam tip: the lecture calls concept maps versatile graphic organizers, but the MCQ key is "**cognitive maps**".

Week 5 · Motivation and Learning

Definition and components

- Motivation is a **"mental impulse that drives and directs human behavior"**; an **internal process that can't be directly observed**, which activates, guides and maintains action
- From the **Latin "movere"** (to move); **Atkinson**: "arousal of tendency to act to produce one or more effects"
- Components: **activation, persistence, intensity, direction** (not "confusion")

Intrinsic vs extrinsic (locus of causality)

- **Intrinsic**: needs, interests, curiosity, self-satisfaction; tied to **deep learning, better performance, self-efficacy**, so more crucial for students
- **Extrinsic**: rewards, **social pressure**, punishment, financial gain; can be turned into intrinsic by changing the task and learning environment
- Continuum: **amotivation** (no intent) → extrinsic → **introjected regulation** (pressure from within) → **identified regulation** (seeing a skill's value) → intrinsic

Drive and incentive theories

- **Drive reduction: Clark Hull, 1943**; behaviour that reduces a drive is strengthened
- **Biological = primary** (food, water, oxygen); **psychological = secondary** (love, attention, approval); **social = tertiary**
- **Incentive theory**: act to gain a reward; based on **operant conditioning**

Exam tip: the slide credits Hull with "three primary drives" (food, sex, social interaction). Love is a psychological drive, not a biological one.

Arousal theory

- People seek an **optimal (not minimal) level of arousal**: low arousal makes them seek stimulation, too much makes them bored and unmotivated
- **Yerkes–Dodson law**: an **inverted-U** curve of arousal against performance

Exam tip: the slide says **Henry Murray (1938)** first proposed it, so answer Murray. Zajonc (1965) and Atkinson (1957) developed it further.

Expectancy theory

- **Vroom, 1964** (extended by Porter and Lawler, 1968): behaviour is a **conscious choice among alternatives**
- **Expectancy**: effort leads to performance ("if I work hard I can hit the targets"); **instrumentality**: performance leads to reward; **valence**: how much the reward is valued (**negative** if you want to avoid it)

Goal setting (Edwin Locke)

- Goals should be specific, challenging but attainable, accepted, with feedback and deadlines
- **SMART**: Specific, **Measurable**, Assignable, Realistic, **Time-based** ("unrealistic" is the trap)

Maslow's hierarchy

- **Five tiers**, from the bottom: **physiological** (food, sleep) → **safety** (job security) → **love/belonging** (friendship) → **esteem** (achievement, education, competence, respect) → **self-actualization**
- Lower needs must be met first; bottom four are **deficiency needs**, the top is a **growth need**

Self-Determination Theory (Ryan & Deci, 2000)

- Three needs: **autonomy, competence, relatedness**
- Separates **autonomous** from **controlled** motivation; intrinsic motivation plus identified regulation are the highest forms

Achievement motivation

- **Atkinson (1966)**: a **need to achieve** and a **need to avoid failure** (worry, self-criticism, faster heart rate)
- **McClelland**: needs for **achievement, affiliation, power**

Week 6 · Emotional Intelligence and Learning

Definitions and founders

- EI is **"the ability to identify and manage one's own emotions and the emotions of others"**, a type of **social intelligence**
- **Salovey & Mayer (1990)** gave the first theory; **Daniel Goleman** popularized it (1995)
- Experts say **EQ is more important than IQ**; it predicts academic and workplace success; EI is **learnable at any stage of life** (not genetic)

Goleman's five areas

- **Self-awareness** (putting words to feelings = **emotional literacy**), **self-regulation** (slowing down impulsive action), **motivation**, **empathy** (how tuned you are to others' emotions), **social skills**
- Empathy and social skills **handle relationships with others**

Signs, advantages, four levels

- Signs: letting go of mistakes, accepting responsibility, curiosity, empathy (not "holding onto mistakes")
- Advantages: accepting criticism, saying no, listening, knowing why we act (not "hiding feelings")
- Four levels: **perceiving** (the first step; verbal and nonverbal cues) → **reasoning with** → **understanding** → **managing** (the highest)
- **Mayer & Salovey (1997)** integrative model, four abilities: perceiving and expressing emotion, using emotion to aid thought, understanding emotion, reflective regulation

Other models

- **Bar-On (2006) mixed model**: competencies, skills and "facilitators"; **10 components**, e.g. self-regard, assertiveness, stress tolerance, impulse control, reality testing
- Facilitators: **self-actualization, independence, social responsibility, optimism, happiness**
- **Nine-layer pyramid**, from the bottom: **emotional stimuli**, emotion recognition, self-awareness, self-management, social awareness, social skills, self-actualization, transcendence, **emotional unity**

Building EI

- Self-awareness: feedback, journalling, meditation, **pursuing one's passions**, positive self-talk
- Self-regulation: cognitive reframing; classroom: "take a break" corner, cooperation chart, writing out conflicts
- Having students write **autobiographies** builds **metacognition**

Emotional labor (Hochschild, 1983)

- Managing feelings to fit **organizational rules**, e.g. **flight attendants**; done mostly by women
- **Feeling rules**: organizations controlling employees' inner feelings
- Managing emotions is a private act shaped by cultural norms and **not directly regulated by others**
- **Surface acting**: change the outward display without feeling it → exhaustion, **burnout**
- **Deep acting**: try to really feel the emotion → a sense of personal accomplishment

Emotional contagion

- One person's emotions trigger similar emotions in others (smile back, angry mobs); based on mimicry and **mirror neurons**
- Negative contagion → more cognitive errors and accidents

Emotional creativity

- Originality and appropriateness in emotional experience; one of the **three areas of general creativity**, with nonverbal/verbal creativity and problem solving
- Linked to openness; women score higher; e.g. **a breakup gives rise to song lyrics**

Week 7 · Learning and Instruction

Definitions of learning

- **Simon**: changes in a system that bring **improved performance** over time on similar tasks
- **Gagne (1982)**: "a change in **human disposition or capacity** that **persists** over time and is **not simply ascribable to growth**"
- **Bloom's taxonomy**: **cognitive** (knowledge), **affective** (feelings), **psychomotor** (physical skills)
- **Dewey**: reflection means **returning to experience** (recalling events) → **attending to feelings** → **evaluating experience**

Five schools

- **Behaviorist**: observable behavior; environment shapes it; **contiguity** and **reinforcement** are central
- **Cognitive**: organized material, prior knowledge, **individual differences**; **cognitive feedback** tells learners of success or failure; reinforcement can be a **knowledge of results** instead of a reward

Exam tip: "individual differences matter" is a cognitive principle, not a behaviorist one.

- **Humanistic (Carl Rogers)**: personal involvement, self-initiated, pervasive; the **locus of evaluation is in the learner**
- **Social/situational (Bandura)**: observation, **situated learning**, communities of practice
- **Constructivist**: the learner builds knowledge

Factors affecting learning

- Learning as a process: **input** → **process** → **output** (certification, attitude change, knowledge enhancement), with two-way feedback
- **Individual**: motivation, interest, perception, learning capability, endurance, **ego**
- **Task**: trainer competence, communication gap, rationality, **task overload**
- **Environmental**: ambience, space, decorum, **ergonomics**, service quality

Learning styles

- **Active learning**: doing, play, experiments; **passive**: listening and memorizing
- **Visual**: maps, charts; **auditory**: sort ideas **after speaking**; **logical**: patterns, causes
- **Social/linguistic**: peer work, **role-play**; **solitary**: study alone; also kinesthetic, reading/writing and nature learners

Instruction

- Instruction is a **set of events outside the learner** that supports the learner's internal learning
- Strategies: **direct** (teacher-centered), **indirect** (student-centered), **interactive** (group), **independent** (individual), **experiential** (inquiry, hands-on)
- **Effective instruction** is interactive, **eclectic**, **generative and dynamic** (not static)
- **Trainer roles**: **expert** (transmits information), instructor, facilitator, co-learner, role model
- **Merrill's principles**: problem-centered, **activation, demonstration, application, integration**
- **ISD** should move at an efficient pace; **ADDIE**: Analysis, Design, Development, **Implementation**, Evaluation
- **Joyce & Weil models**: information processing, behavioral, social interaction, personal

Instructional theorists

- **Ausubel**: **meaningful learning** links new knowledge to existing schemas; uses **advance organizers** (**comparative** ones recall old schemas, **expository** ones introduce unfamiliar material) and subsumption
- **Gagne**: five outcome types (verbal information, intellectual skills, cognitive strategies, motor skills, attitudes); **nine events of instruction**, from **gaining attention** to enhancing retention and transfer
- **Bruner**: guided discovery learning and **scaffolding**; **enactive** (0–1 yr) → **iconic** (1–6) → **symbolic** (7+)
- **Piaget's learning cycle**: exploration → concept introduction → concept application

Week 8 · Learning and Pedagogy

Pedagogy

- A **method or way of teaching**; from the **Greek "pedagogue"**, **"the art of teaching children"**
- It improves teaching quality and **enhances the student-teacher relationship**
- Good pedagogy is about **eliciting responses that demonstrate understanding**; it is **student-centered** ("facilitator-centered" is false)
- Other forms: **social**, **culturally responsive** (respects diversity), **personalizing** (online choice), **critical** (social justice; unlearning → learning → relearning → reflecting → evaluating)

Five approaches

- **Constructivist**: "learners are the **makers of meaning and knowledge**"
- **Collaborative**: members interact, share experiences and take **asymmetry roles**
- **Inquiry-based**: **active** learning that starts by **posing questions**; includes problem-based learning; types: confirmation, **structured, guided, open**
- **Integrative**: three steps, **integrative inquiry, application and transfer, reflection**
- **Reflective**: teachers analyze and improve their own practice

Exam tip: "collective", "refractive" and "intuitive" are traps; none of them is an approach.

- **Pedagogical Model**: describes what effective teachers do; **five phases**: engage, explore, explain, elaborate, evaluate

Experiential learning

- Students **take initiative, make decisions, are accountable**; engaged **intellectually, emotionally, socially, soulfully, physically**
- Instructor's roles: setting experiences, **posing problems**, **setting boundaries**, supporting students
- Process: experiencing → sharing → **processing ("What's important?")** → generalizing → applying

Kolb (1984)

- **Four-stage cycle**: concrete experience → reflective observation → abstract conceptualization → **active experimentation (planning)**
- Two continuums: **perception** and **processing**
- **Diverging** (watch rather than do); **assimilating** (logical); **converging** (technical tasks, left alone); **accommodating** (hands-on, **gut feeling**)
- Factors that shape a style: career choice, personality, specialization (**not genes**)

Mayer's multimedia learning

- Assumptions: **dual channels**, **limited capacity**, **active processing**; draws on Sweller's cognitive load theory
- **Multimedia**: words plus pictures beat words alone; **spatial contiguity**: words placed near pictures; **temporal contiguity**: words and pictures shown **simultaneously**
- **Coherence**: leave out extra material; **modality**: **animation + narration** beats animation + on-screen text; **redundancy**: don't add on-screen text to narration

Blended and flipped learning

- **Blended**: online materials plus face-to-face class; **flipped**: content studied at home, higher-order work done in class
- **Blended flip**: Bergmann & Sams; videos of 20–25 minutes at most

Transformative learning

- **Frame of reference = habits of mind + point of view**
- **Mezirow (1978)**: **ten steps**, from the **disorienting dilemma** to **reintegration**
- Sterling's third-order (**epistemic**) learning changes the learner's worldview
- **Contemplative pedagogy**: mindfulness, reflection, compassion

Week 9 · Psychology of E-Learning

Definition, features, types

- **E-learning** = "any form of **teaching, training or tutoring**" for learners of any age, via internet/mobile, using electronic multimedia; implies **computer-enhanced learning**
- **Features**: accessibility, **user friendliness**, **social justice**; personalization, individualization, **learner-centeredness**; open and lifelong learning; bridging the **digital divide**
- **Digital classroom** = "the changing face of learning"; students value **"how to"** over "what"
- E-learning is **NOT a synonym** of audio-visual, multimedia or distance learning (only complementary)
- Types: **text-driven** (text, images, audio, PPT, text questions), **interactive** (graphics, charts, videos), **simulation** (highly interactive, incl. games)
- Benefits: accommodates everyone (worker, housewife, student), lectures repeatable, **scalability** (new training policies), quick delivery, consistency, reduced cost, paperless

Exam tip: teacher-centeredness, stereotypical, closed learning, physical type are NOT e-learning answers.

Improving e-learning

- Cognitive psychology places the **individual** at the centre; "most powerful learner variable is **prior knowledge**"
- Learning is: based on prior knowledge; **idiosyncratic** (no two learners share experiences); **goal-oriented**; depends on **self-awareness** (good learners know **what, why and how**); **social** (e-mail, online discussions, virtual communities); **active** ("crucial determinant of successful e-learning")
- Brain: **36,000 visual cues per hour**; **~85 billion neurons**; practice builds **stronger connections between neurons**
- Process: perception, attention, memory, retrieval

Risks and design

- **Cognitive load theory**: overloading **working memory** blocks new learning; multitasking gives **shorter attention spans**; use headings, simple language, short videos; handwritten notes aid **encoding**
- **Five critical factors**: motivation, emotion, self-discipline, social skills, stress
- Usability: **learnability** (speed and ease for a novice to reach proficiency), efficiency, memorability, errors, satisfaction
- **Topic map** = navigable index; reduces cognitive load
- **Scenario-based/immersive** e-learning: simulated job-related situations in which learners judge and make choices
- Behaviourist ID: discrimination, generalization, association, **chaining** (automatic performance of predetermined steps, drills)
- **Knowles** andragogy: 5 assumptions (self-concept, experience, readiness, orientation, motivation; 4 in 1980, 5th added 1984) and 4 principles (involvement, experience, relevance, problem-centred)
- **ADDIE** = Analysis, Design, Develop, Implement, Evaluate; **Merrill 2002**: task-centred, activation, demonstration, application, integration; **Gagne** 9 events (gain attention first, retention and transfer last)
- **Bloom 1956**, revised by **Anderson and Krathwohl 2001** (Creating on top); **Mayer**: multimedia learning
- **Concepts**, **prototypes** (most recognizable example), **schemas** (assimilation broadens, accommodation creates new)

Week 10 · Effective Learning: Assessment and Feedback

New normal and teachers

- Covid: **1.5 billion** learners moved to **remote learning**
- **Readiness Model** for New Normal: parents', teachers', learners' readiness; most critical factor = **readiness of participants**
- Learner readiness training: **netiquette, time management, study skills**
- Brainly: 53.3% prefer **blended**; 38.7% happy to rejoin school
- Teacher: **dispenser** of knowledge replaced by **facilitator**
- **GESF 2018**, 5 skills: strong subject knowledge, being a facilitator, strong social skills, **external focus**, data analysis
- Needs **digital skills, life skills, communication**; participatory education demands **creativity**

20 psychological principles

- Beliefs about intelligence affect learning: **malleable** (growth mindset) learners focus on goals despite obstacles
- Prior knowledge affects learning; development **NOT limited** by general stages; learning is **context-based** (transfer must be facilitated); long-term learning depends on **practice**
- **Clear, explanatory and timely feedback**; self-regulation can be taught; creativity can be encouraged
- **Intrinsic > extrinsic** motivation; **mastery goals** (acquire/improve skills) > performance goals
- Teacher expectations matter (**Pygmalion**, Rosenthal and Jacobson)
- **Proximal** (short-term), specific, challenging goals beat **distal** (long-term) goals
- Learning in multiple social contexts; interpersonal relationships; **emotional well-being**
- Classroom management: high expectations, positive relationships, high support
- **Formative and summative** both useful, interpreted differently

Exam tip: "cognitive development is limited by stages" is the NOT-a-principle trap.

Five principles and evaluation

- **Individual differences, motivation, human development, reward, punishment**
- Individual differences handled by observation, interviews, rating scales, questionnaires, **sociograms** (never ignorance)
- Reward: **social** (clapping) or **tangible** (money, book)
- **Evaluation** = "process of judging the value or worth of an individual's achievements or characteristics"
- **Formative** = assessment FOR learning; "frequent, interactive assessment of student progress" (OECD 2005); vast majority **informal**; anecdotal records, quizzes, essays, lab reports
- **Summative** = assessment OF learning; **after instruction**; unit tests, final exams, **entrance exams**
- **Assessment for Learning** = teaching approach generating feedback students use to improve
- Effective evaluation: **alignment, validity, reliability, transparency**

Feedback

- Feedback attached to a grade loses benefit (Black and Wiliam)
- **7 principles of good feedback**: self-assessment, dialogue, clarify good performance, close gap, high-quality information, positive motivation and self-esteem, inform teaching
- Must be **constructive, timely, meaningful**
- Stages: **feed-up** (where am I going?), **feedback** (how am I doing?), **feed-forward** (where next?)
- Levels: **task**, **process**, **personal** ("That's an intelligent response, well done")
- **Natriello 1987** 8-phase cycle; **Narciss** ITFL (interactive two-feedback-loops) model

Exam tip: "facilitator-level" feedback does not exist.

Week 11 · Learning and Social Cognition

Social cognition

- "Complex set of mental abilities underlying **social stimulus perception, processing, interpretation and response**"; cognitive processes influencing **social behavior**
- **Protracted development from infancy to adulthood**; develops in childhood and adolescence, NOT old age
- Recruits perception, attention, language, memory, executive functions
- Basic: **face processing, joint attention**; complex: **theory of mind, moral reasoning**, social decision-making
- **Hot** = affective (facial expressions); **cold** = cognitive (beliefs, intentions)
- **Social schemas** = mental representations of social **patterns and norms**
- Young children are **egocentric** (Piaget); **theory of mind** = understanding others' **mental states**
- Culture matters: two people may interpret one situation differently

Exam tip: rote memory is NOT a social-cognition ability.

Bandura's social cognitive theory

- **Albert Bandura** (Stanford); an extension of social learning theory that adds cognition (judgment, motivation)
- Covers **observational learning, modeling, self-efficacy**; people are **active agents**; central tenet: a sense of **agency**
- Observational learning, **4 processes**: **attentional** (what is selected), **retention** (aided by rehearsal), **production**, **motivational** (rewarded behaviour is reproduced, punished behaviour is not)
- **Learning without performance**: people do not perform everything they learn
- Children observing aggressive models become more aggressive; moral judgments also develop partly through modeling
- Modeling teaches new behaviours, changes how often learned ones occur, and may encourage forbidden ones
- Models: **live** (actual person), **symbolic** (characters in books, films, TV, online), **verbal instructional**
- **Reciprocal determinism**: behavior, personal and environmental factors influence each other
- **Reinforcements** = internal/external responses affecting continuing a behavior; **expectations** = anticipated consequences
- **Self-efficacy** = confidence in performing a behavior; unique to SCT (the Theory of Planned Behavior added it later)
- Social-cognitive topics: attitudes, person perception, prejudice, stereotypes, self-concept, discrimination, persuasion, decision-making

Digital learning, self-efficacy, Johari

- **CASTLE** hypotheses: activation, social cue strength, **cognitive influence** (social processes shape selection, organization, integration, retrieval), interaction, schema influence, social mediation
- **Social facilitation**: social presence improves performance; **pedagogical agents** = computer-generated tutors
- Piaget's **cognitive conflict**: inconsistent information triggers assimilation/accommodation
- Self-efficacy is task-specific; self-esteem = self-worth; self-confidence = general
- Self-efficacy sources: **mastery experiences** (strongest), social modeling, social persuasion, psychological responses; opposite = **learned helplessness**
- Social perception: **observation, attribution, integration, confirmation**
- **Johari Window** (Luft and Ingham, 1955): **arena**, **facade**, **blind spot**, **unknown**; grow arena via **self-disclosure** and **feedback**
- Social learning types: **facilitation** and **imitation**

Week 12 · Learning for Sustainability

Concepts and history

- Environmental, economic, social **sustainability** = **unidimensional**; **sustainable development** = **multidimensional** (Eggert 2009)

Exam tip: the lecture follows Eggert, so answer "unidimensional" for environmental or economic sustainability.

- Term first in a book for the **1972 UN Stockholm conference on human environment**
- **Brundtland Commission**, Our Common Future, **1987**: SD solves **environmental degradation**; "meets the needs of the present without compromising the ability of future generations to meet their own needs"
- Pillars: environment, economy, society
- Environmental impacts on air, water, land, **biodiversity**, human health, ocean
- Social: labour practices, human rights, business ethics, safety
- Economic: productivity, **revenue generation**, resource use, jobs
- **Social entrepreneurship** = solving social problems with **market-based methods**

SDGs and SDG 4

- 17 SDGs (2015, target 2030); key ones: 4 quality education, 6 **clean water and sanitation**, 12 responsible consumption, 16 **peace and justice**, 17 partnerships
- **Education is related to all SDGs**
- **SDG 4**: "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all"
- **7 targets**: 4.1 **universal primary and secondary**, 4.2 early childhood, 4.3 TVET/higher, 4.4 skills for decent work, 4.5 gender equality and inclusion, 4.6 youth literacy, 4.7 citizenship/ESD

- **3 means**: learning environments, **scholarships**, teachers

Exam tip: "privatization of institutes" is NOT an SDG 4 target.

- **HESI** = **Higher Education Sustainability Initiative**: role of **higher education**; launched **20 Oct 2020**, day 1 of the **Global Education Meeting**

ESD

- **GAP** = **Global Action Programme**, succeeded by **ESD for 2030**
- 5 priority areas: policy, learning environments, educators, youth, local action
- ESD for 2030 targets **everyone**: policymakers, institutional leaders, educators, learners, **parents**, youth, communities
- **Okayama Commitment 2014**: responding, engaging, enabling, embedding, sustaining, transforming
- ESD is "holistic and transformational"; needs a shift **from teaching to learning**
- **8 competencies**: systems thinking, **anticipatory** (multiple futures), normative, strategic, collaboration, critical thinking, **self-awareness** (own role in community and society), integrated problem-solving
- UNESCO's 3 learning dimensions: **cognitive, social and emotional, behavioural**

Exam tip: "biological learning" is the trap option.

- **Transformative learning** = changing deeply held assumptions; most prominent theory for SD
- **Mohanty 2019** model: profit (MHRD policy), people (human resources), planet (educational ecosystem)
- **2021 Cedefop** skills forecast for the **European Green Deal**; **GreenComp** = 4 areas, 3 competences each